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Stochastic Processes

Lecture 22

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Bayesian Logistic Regression using Challenger O-Ring Dataset

- Between 1981 and 1985, NASA launched 23 space-shuttle missions before the ill-fated Challenger flight on 28 Jan 1986
- For every flight $i \in \{1, \ldots, 23\}$, engineers recorded
  - the ambient air-temperature at lift-off x_i (in °F), and
  - $y_i = \mathbf{1}\{\text{whether at least one solid-rocket-booster O-ring was damaged during ascent}\}$
- The lowest temperature available in the dataset was 53°F, whereas Challenger launched at 31°F despite warnings from engineers

Objective

Deduce the functional relationship between ambient temperature at lift-off and probability of ring damage

Tool: **Logistic regression**

Regression Model

- For data point i, let
$$\eta_i = \beta_0 + \beta_1 x_i.$$
- Taking $\boldsymbol{\beta} = (\beta_0, \beta_1)$, let
$$\pi_i \;=\; \mathrm{Logit}(\eta_i) \;=\; \frac{\exp \eta_i}{1 + \exp \eta_i}.$$
- Conditional distribution of output, given input and parameters:
$$\mathbb{P}(y_i = 1 \mid x_i, \boldsymbol{\beta}) \;=\; \pi_i \;=\; 1 - \mathbb{P}(y_i = 0 \mid x_i, \boldsymbol{\beta}).$$
- Data points $\{(x_i, y_i)\}_{i=1}^{23}$ are IID

Prior, Likelihood, and Posterior

- $\boldsymbol{\beta} = (\beta_0, \beta_1)$ unknown; a reasonable prior is

$$\beta_0, \beta_1 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad g(\boldsymbol{\beta}) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\beta_0^2 + \beta_1^2}{2\sigma^2}\right).$$

- Likelihood function is given by

$$p(\mathbf{y} \mid \mathbf{x}, \beta) = \prod_{i=1}^{23} p(y_i \mid x_i, \beta) = \prod_{i=1}^{23} \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

- Posterior function is given by

$$f(\boldsymbol{\beta} \mid \mathbf{x}, \mathbf{y}) = \frac{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta})\, g(\boldsymbol{\beta})}{C(\mathbf{x}, \mathbf{y})}, \qquad C(\mathbf{x}, \mathbf{y}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}')\, g(\boldsymbol{\beta}')\, \mathrm{d}\beta_0'\, \mathrm{d}\beta_1'.$$

- **No closed-form expression for $C(\mathbf{x}, \mathbf{y})$**
  Posterior is known only up to a constant
- Can we still sample from the posterior? Yes, via Metropolis–Hastings!

Setting up Metropolis–Hastings

- Set Q matrix (kernel) as
$$Q(\boldsymbol{\beta}, \cdot) = \mathcal{N}(\boldsymbol{\beta}, I),$$
where I denotes the 2×2 identity matrix (**easy to sample from Q**)
- Q is symmetric
- The acceptance probabilities (using $S = S^{(M)}$ from previous lecture) are then given by

$$A(\boldsymbol{\beta}, \boldsymbol{\beta}') = \min\left\{1, \frac{f(\boldsymbol{\beta}' \mid \mathbf{x}, \mathbf{y}) \cdot Q(\boldsymbol{\beta}', \beta)}{f(\boldsymbol{\beta} \mid \mathbf{x}, \mathbf{y}) \cdot Q(\boldsymbol{\beta}, \boldsymbol{\beta}')}\right\} = \min\left\{1, \frac{p(\mathbf{y} \mid \mathbf{x}, \beta') \cdot g(\beta')}{p(\mathbf{y} \mid \mathbf{x}, \beta) \cdot g(\beta)}\right\}.$$

Metropolis–Hastings

- For any bounded, continuous function h, we have by the ergodic theorem that

$$\frac{1}{n}\sum_{k=1}^{n} h(\boldsymbol{\beta}_k) \quad \xrightarrow{\text{a.s.}} \quad \int h(\boldsymbol{\beta}')\, f(\boldsymbol{\beta}' \mid \mathbf{x}, \mathbf{y})\, \mathrm{d}\boldsymbol{\beta}',$$

  where $\{\boldsymbol{\beta}_k\}_{k\in\mathbb{N}}$ is the ergodic DTMC generated by the Metropolis–Hastings algorithm (with the prior $g(\cdot)$ as its initial distribution and the posterior $f(\cdot \mid \mathbf{x}, \mathbf{y})$ as its stationary distribution)
- What if we are just interested in one sample $\quad \boldsymbol{\beta} \sim f(\cdot \mid \mathbf{x}, \mathbf{y})$?
- Because $\{\boldsymbol{\beta}_k\}_{k\in\mathbb{N}}$ is ergodic, we know that

$$\lim_{n\to\infty} \left\| \mu_n - f(\cdot \mid \mathbf{x}, \mathbf{y}) \right\|_{\mathrm{TV}} \quad \longrightarrow \quad 0,$$

  where μ_n is the distribution of $\boldsymbol{\beta}_n$
- We run the Metropolis–Hastings algorithm until μ_n is "close" to the target $f(\cdot \mid \mathbf{x}, \mathbf{y})$ (e.g., TV distance is below a desired threshold ε)

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Rate of Convergence of TV for Metropolis–Hastings

From the paper [MT96]

RATES OF CONVERGENCE OF THE HASTINGS AND METROPOLIS ALGORITHMS[1]

BY K. L. MENGERSEN AND R. L. TWEEDIE

Queensland University of Technology and Colorado State University

We apply recent results in Markov chain theory to Hastings and Metropolis algorithms with either independent or symmetric candidate distributions, and provide necessary and sufficient conditions for the algorithms to converge at a geometric rate to a prescribed distribution π. In the independence case (in $\mathbb{R}^k$) these indicate that geometric convergence essentially occurs if and only if the candidate density is bounded below by a multiple of π; in the symmetric case (in $\mathbb{R}$ only) we show geometric convergence essentially occurs if and only if π has geometric tails. We also evaluate recently developed computable bounds on the rates of convergence in this context: examples show that these theoretical bounds can be inherently extremely conservative, although when the chain is stochastically monotone the bounds may well be effective.

1. Hastings and Metropolis algorithms and Markov chains. The Hastings [6] and Metropolis [9] algorithms allow simulation of a probability density $\pi(x)$ which is only known up to a factor, that is, when only $\pi(x)/\pi(y)$ is known. This is especially relevant when π is the posterior distribution in a Bayesian context: see [3], [21], [20], [13], [22] and [4] for a more detailed introduction.

In this paper we are concerned with the rate of convergence of these algorithms for distributions on $\mathbb{R}^k$, and in particular with finding conditions under which the convergence is geometrically fast.

In order to describe the Hastings and Metropolis algorithms, we first consider a *candidate transition kernel* $Q(x, A)$, $x \in \mathbb{R}^k$, $A \in \mathscr{B}(\mathsf{X})$, satisfying

$$Q(x, A) \geq 0, \qquad Q(x, \mathbb{R}^k) = 1,$$

which generates potential transitions for a discrete-time Markov chain evolving on $\mathsf{X} = \mathbb{R}^k$ equipped with the Borel σ-algebra $\mathscr{B}(\mathsf{X})$. We usually assume that $Q(x, \cdot)$ is absolutely continuous with density $q(x, y)$ with respect to Lebesgue measure μ^{Leb}, except perhaps for an atom $Q(x, \{x\}) > 0$, although this is not necessary and the results below will hold in more general contexts also.

Rate of Convergence of TV for Metropolis–Hastings

From the paper [MT96]

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THEOREM 1.3. *For any Markov chain* Φ *the following are equivalent*:

(i) *The chain is aperiodic and Doeblin's condition holds; that is, there is a probability measure* ϕ *on* $\mathscr{B}(\mathsf{X})$ *and* $\varepsilon < 1$, $\delta > 0$, $m \in \mathbb{Z}_+$ *such that, whenever* $\phi(A) > \varepsilon$,

(11) $$\inf_{x \in \mathsf{X}} P^m(x, A) > \delta.$$

(ii) *The whole state space* X *is small; that is, for some* $\delta > 0$ *and some* $m \geq 1$ *and some probability measure* ν,

(12) $$P^m(x, A) \geq \delta\nu(A), \qquad x \in \mathsf{X},\ A \in \mathscr{B}(\mathsf{X}).$$

(iii) *For some small set* C *we have*

$$\sup_{x \in \mathsf{X}} \mathsf{E}_x[\tau_C] < \infty,$$

where τ_C *is the first return time to* C.

(iv) *The chain* Φ *is uniformly ergodic*.

When any of (i)–(iv) *hold the convergence in* (10) *is geometrically fast*, *and for any* x *we can bound the rate of convergence using* (12) *by*

(13) $$\|P^n(x, \cdot) - \pi\| \leq (1 - \delta)^{[n/m]}.$$

Gibbs Sampling

(A Special Case of Metropolis–Hastings Algorithm)

Gibbs Sampling as a Special Case of Metropolis–Hastings

- Let the target distribution be a PMF/density π on

$$\mathcal{X} \subseteq \mathbb{R}^d, \qquad \mathbf{x} = (x_1, \ldots, x_d).$$

- Fix $i \in \{1, \ldots, d\}$
  Define the **proposal kernel** $Q = Q_i$ by

$$Q_i(\mathbf{x}, \mathbf{y}) = \mathbf{1}\{\mathbf{y}_{-i} = \mathbf{x}_{-i}\}\, \pi(y_i \mid \mathbf{x}_{-i}),$$

  where $\mathbf{x}_{-i}$ denotes all coordinates of $\mathbf{x}$ except x_i
- Taking $S = S^{(M)}$ as the choice of the symmetric S matrix (see previous lecture), the corresponding acceptance matrix $A^{(M)}$ is given by

$$A^{(M)}(\mathbf{x}, \mathbf{y}) \;=\; \min\left\{1,\; \frac{\pi_{\mathbf{y}}\, Q_i(\mathbf{y}, \mathbf{x})}{\pi_{\mathbf{x}}\, Q_i(\mathbf{x}, \mathbf{y})}\right\}.$$

- **Exercise:** It is easy to show that

$$\pi_{\mathbf{y}}\, Q_i(\mathbf{y}, \mathbf{x}) \;=\; \pi_{\mathbf{x}}\, Q_i(\mathbf{x}, \mathbf{y}) \qquad \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{X}.$$

  From this, it follows that $A^{(M)}(\mathbf{x}, \mathbf{y}) = 1$ for all $\mathbf{x}, \mathbf{y}$

Gibbs Sampling as a Special Case of Metropolis–Hastings

- **The Gibbs sampling algorithm:**
  - Start with an arbitrary $\mathbf{X}^{(1)} = \mathbf{x}$
  - Sample $\mathbf{Y}$ as follows: let $\mathbf{Y}_{-i} = \mathbf{x}_{-i}$, and let
  $$Y_i \sim \pi(\cdot \mid \mathbf{x}_{-i}) = \pi(\cdot \mid \mathbf{X}^{(1)}_{-i}).$$
  - Set $\mathbf{X}^{(2)} = \mathbf{Y}$
  - Sample $\mathbf{Y}$ as follows: let $\mathbf{Y}_{-i} = \mathbf{X}^{(2)}_{-i}$, and let
  $$Y_i \sim \pi(\cdot \mid \mathbf{X}^{(2)}_{-i}).$$
  - Set $\mathbf{X}^{(3)} = \mathbf{Y}$, and continue in this fashion
- In practice, one coordinate i is not fixed, but is itself **chosen uniformly at random**
  In that case, the proposal kernel Q becomes
  $$Q(\mathbf{x}, \mathbf{y}) = \frac{1}{n} \sum_{i=1}^{n} Q_i(\mathbf{x}, \mathbf{y}),$$
  where Q_i is as defined earlier
- It can be shown that Q satisfies $\pi_{\mathbf{y}}\, Q(\mathbf{y}, \mathbf{x}) = \pi_{\mathbf{x}}\, Q(\mathbf{x}, \mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$

Some Useful References on Markov Chains

- The paper [RR04] presents convergence rates for TV distance under Gibbs sampling
- A comprehensive treatment of continuous-state Markov chains may be found in [MT12]
- A coverage of the notion of lumpability and conditions under which functions of Markov chains are Markov chains may be found in [RS91] and [KS+69]

Tidbit: Test for Irreducibility

Theorem (Test for Irreducibility)

A row stochastic matrix P of size $d \times d$ is irreducible if and only if

$$\sum_{k=0}^{d-1} P^k > \mathbf{0},$$

where $P^0 = I$, $\mathbf{0}$ denotes the all-zeros matrix of size $d \times d$, and the inequality is entry-wise.

A simple algorithm to check for irreducibility:

- Start with $M_0 = I$.
- Recursively define
$$M_{\ell+1} = I + PM_\ell.$$
- If $M_\ell > \mathbf{0}$ for any $1 \leq \ell \leq d-1$, then P is irreducible

Conditional Expectations

Reference book: Durrett [Dur19]

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Conditional Expectation

Fix a probability space $(\Omega, \mathscr{F}_0, \mathbb{P})$.

Definition (Conditional Expectation)

Let $\mathscr{F} \subset \mathscr{F}_0$ be a sub-σ-algebra of $\mathscr{F}_0$.
Let $X : \Omega \to \mathbb{R}$ be measurable with respect to $\mathscr{F}$ **with** $\mathbb{E}\big[|X|\big] < +\infty$.
The conditional expectation of X given $\mathscr{F}$, denoted $\mathbb{E}[X \mid \mathscr{F}]$, is **any random variable** Y satisfying:

1. Y is measurable with respect to $\mathscr{F}$.
2. For all $A \in \mathscr{F}$, we have

$$\mathbb{E}[X\,\mathbf{1}_A] = \int_A X \,\mathrm{d}\mathbb{P} = \int_A Y \,\mathrm{d}\mathbb{P} = \mathbb{E}[Y\,\mathbf{1}_A].$$

Two immediate questions:

- Given $\mathscr{F}$ and X which is measurable w.r.t. $\mathscr{F}$, does there exist at least one Y satisfying 1, 2 above?
- If there exists at least one Y, is it unique?

An Important Result from Measure Theory

Theorem (Radon–Nikodym Theorem)

Fix a measurable space $(\Omega, \mathscr{F})$. Let μ, ν be two measures on $(\Omega, \mathscr{F})$.
Let $\nu \ll \mu$, i.e., $\mu(A) = 0$ implies $\nu(A) = 0$.
Then there exists a **non-negative, $\mathscr{F}$-measurable function** $f : \Omega \to [0, +\infty)$ such that

$$\nu(A) = \int_A f \, \mathrm{d}\mathbb{P} \qquad \forall\, A \in \mathscr{F}.$$

- The function f is called the **Radon–Nikodym derivative** of μ with respect to ν, and is denoted $\dfrac{\mathrm{d}\mu}{\mathrm{d}\nu}$

Important

PDF in probability theory has its roots in the Radon–Nikodym Theorem of measure theory.

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Existence

- Given $\mathscr{F}$ and X which is **non-negative** and measurable w.r.t. $\mathscr{F}$, define

$$\nu(A) = \int_A X \, \mathrm{d}\mathbb{P} = \mathbb{E}[X \mathbf{1}_A], \qquad A \in \mathscr{F}.$$

- ν satisfies the following two properties:
  - $\nu(\Omega) = \int_\Omega X \, \mathrm{d}\mathbb{P} = \mathbb{E}[X] < +\infty.$
  - For any disjoint $A_1, A_2, \ldots$, we have

$$\nu\left(\bigsqcup_{n\in\mathbb{N}} A_n\right) = \mathbb{E}\left[X \mathbf{1}_{\bigsqcup_{n\in\mathbb{N}} A_n}\right] = \mathbb{E}\left[\sum_{n\in\mathbb{N}} X \mathbf{1}_{A_n}\right] \overset{\text{MCT}}{=} \sum_{n\in\mathbb{N}} \mathbb{E}[X \mathbf{1}_{A_n}] = \sum_{n\in\mathbb{N}} \nu(A_n).$$

  Thus, ν is a finite measure on $(\Omega, \mathscr{F})$, and $\nu \ll \mathbb{P}$

- By the Radon–Nikodym theorem, there exists a non-negative, $\mathscr{F}$-measurable function $Y : \Omega \to [0, +\infty)$ such that

$$\nu(A) = \int_A Y \, \mathrm{d}\mathbb{P} \qquad \forall\, A \in \mathscr{F}.$$

  This Y is denoted $\mathbb{E}[X \mid \mathscr{F}]$

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